



Stony Brook Institute
at Anhui University

安徽大学纽约石溪学院

CSE 306 Operating Systems

02: Operating-System Structures

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Outline

- Operating System Services
- User and Operating System-Interface
- System Calls
- System Services
- Linkers and Loaders
- Why Applications are Operating System Specific
- Design and Implementation
- Operating System Structure
- Building and Booting an Operating System
- Operating System Debugging

Objectives

- Identify services provided by an operating system
- Illustrate how system calls are used to provide operating system services
- Compare and contrast monolithic, layered, microkernel, modular, and hybrid strategies for designing operating systems
- Illustrate the process for booting an operating system
- Apply tools for monitoring operating system performance
- Design and implement kernel modules for interacting with a Linux kernel

Operating System Services

- Operating systems provide an environment for execution of programs and services to programs and users
- One set of operating-system services provides functions that are helpful to the user:
 - **User interface** - Almost all operating systems have a user interface (**UI**).
 - ▶ Varies between **Command-Line (CLI)**, **Graphics User Interface (GUI)**, **touch-screen**, **Batch**
 - **Program execution** - The system must be able to load a program into memory and to run that program, end execution, either normally or abnormally (indicating error)
 - **I/O operations** - A running program may require I/O, which may involve a file or an I/O device
 - **File-system manipulation** - The file system is of particular interest. Programs need to read and write files and directories, create and delete them, search them, list file Information, permission management.

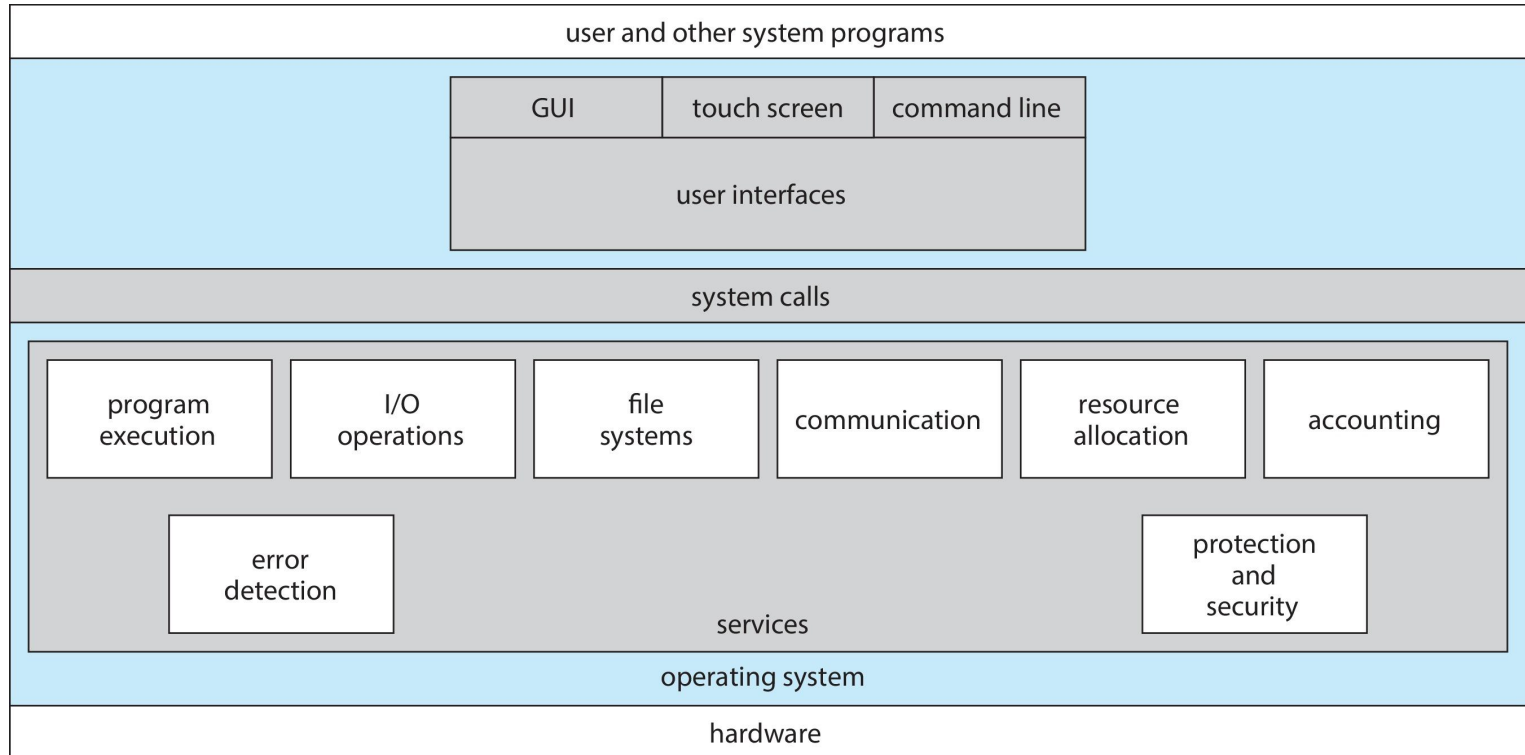
Operating System Services

- One set of operating-system services provides functions that are helpful to the user
 - **Communications** – Processes may exchange information, on the same computer or between computers over a network
 - ▶ Communications may be via shared memory or through message passing (packets moved by the OS)
 - **Error detection** – OS needs to be constantly aware of possible errors
 - ▶ May occur in the CPU and memory hardware, in I/O devices, in user program
 - ▶ For each type of error, OS should take the appropriate action to ensure correct and consistent computing
 - ▶ Debugging facilities can greatly enhance the user's and programmer's abilities to efficiently use the system

Operating System Services

- Another set of OS functions exists for ensuring the efficient operation of the system itself via resource sharing
 - **Resource allocation** - When multiple users or multiple jobs running concurrently, resources must be allocated to each of them
 - ▶ Many types of resources - CPU cycles, main memory, file storage, I/O devices.
 - **Logging** - To keep track of which users use how much and what kinds of computer resources
 - **Protection and security** - The owners of information stored in a multiuser or networked computer system may want to control use of that information, concurrent processes should not interfere with each other
 - ▶ **Protection** involves ensuring that all access to system resources is controlled
 - ▶ **Security** of the system from outsiders requires user authentication, extends to defending external I/O devices from invalid access attempts

A View of Operating System Services



Command Line interpreter

- CLI allows direct command entry
- Sometimes implemented in kernel, sometimes by systems program
- Sometimes multiple flavors implemented – **shells**
- Primarily fetches a command from user and executes it
- Sometimes commands built-in, sometimes just names of programs
 - If the latter, adding new features doesn't require shell modification

Bourne Shell Command Interpreter

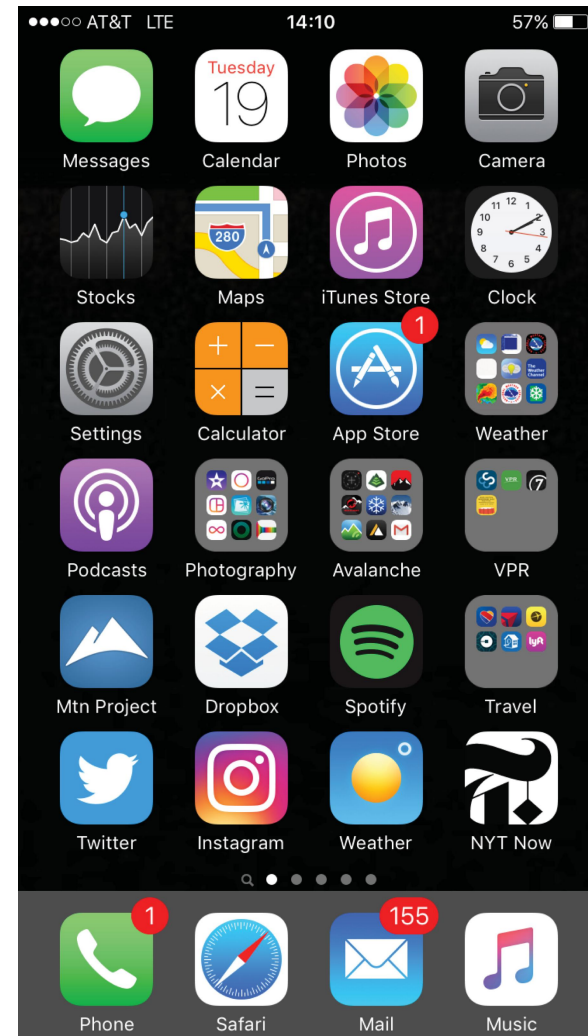
```
1. root@r6181-d5-us01:~ (ssh)
root@r6181-d5-us01:~ (ssh)
Last login: Thu Jul 14 08:47:01 on ttys002
iMacPro:~ pbg$ ssh root@r6181-d5-us01
root@r6181-d5-us01's password:
Last login: Thu Jul 14 06:01:11 2016 from 172.16.16.162
[root@r6181-d5-us01 ~]# uptime
 06:57:48 up 16 days, 10:52,  3 users,  load average: 129.52, 80.33, 56.55
[root@r6181-d5-us01 ~]# df -kh
Filesystem                Size      Used Avail Use% Mounted on
/dev/mapper/vg_ks-lv_root   50G       19G   28G  41% /
tmpfs                      127G      520K   127G   1% /dev/shm
/dev/sda1                   477M       71M   381M  16% /boot
/dev/dssd0000               1.0T     480G   545G  47% /dssd_xfs
tcp://192.168.150.1:3334/orangefs 12T     5.7T   6.4T  47% /mnt/orangefs
/dev/gpfs-test              23T     1.1T   22T   5% /mnt/gpfs
[root@r6181-d5-us01 ~]#
[root@r6181-d5-us01 ~]# ps aux | sort -nrk 3,3 | head -n 5
root      97653 11.2  6.6 42665344 17520636 ?    S<Ll  Jul13 166:23 /usr/lpp/mmfs/bin/mmfsd
root      69849  6.6  0.0      0      0 ?        S    Jul12 181:54 [vpthread-1-1]
root      69850  6.4  0.0      0      0 ?        S    Jul12 177:42 [vpthread-1-2]
root       3829  3.0  0.0      0      0 ?        S    Jun27 730:04 [rp_thread 7:0]
root       3826  3.0  0.0      0      0 ?        S    Jun27 728:08 [rp_thread 6:0]
[root@r6181-d5-us01 ~]# ls -l /usr/lpp/mmfs/bin/mmfsd
-r-x----- 1 root root 20667161 Jun  3  2015 /usr/lpp/mmfs/bin/mmfsd
[root@r6181-d5-us01 ~]#
```

User Operating System Interface - GUI

- User-friendly **desktop** metaphor interface
 - Usually mouse, keyboard, and monitor
 - **Icons** represent files, programs, actions, etc
 - Various mouse buttons over objects in the interface cause various actions (provide information, options, execute function, open directory (known as a **folder**))
 - Invented at Xerox PARC
- Many systems now include both CLI and GUI interfaces
 - Microsoft Windows is GUI with CLI “command” shell
 - Apple Mac OS X is “Aqua” GUI interface with UNIX kernel underneath and shells available
 - Unix and Linux have CLI with optional GUI interfaces (CDE, KDE, GNOME)

Touchscreen Interfaces

- Touchscreen devices require new interfaces
 - Mouse not possible or not desired
 - Actions and selection based on gestures
 - Virtual keyboard for text entry
- Voice commands



A collage of various computer screens and windows, illustrating a multi-tasking environment. The top row shows a PowerPoint window with a presentation slide titled 'DSSD HIGH PERFORMANCE PARALLEL FILE SYSTEMS', a file explorer window showing a directory structure, and a dictionary window. The middle row features a presentation slide with a technical diagram, a file explorer window, and a calculator window. The bottom row displays a presentation slide with a technical diagram, a file explorer window, and a calculator window. The overall theme is productivity and multitasking in a professional or academic setting.